

A
Project Synopsis
On

“Predicting Housing Prices”

Submitted in partial fulfilment of requirement for the award of B.TECH in Computer
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CERTIFICATE

This is to certify that the project report on **“Predicting Housing Prices”**

Submitted by **“Shubham Landge”** of **Seventh Sem**, AI&DSY Dept. in satisfactory manner for the partial fulfilment of the requirement of **“AI&DSY Engineering”** offered by DBATU, Lonere. For **Academic Year 2024-2025**.

It is an original work carried under the supervision and guidance of **“Shazia Shaheen Ansari”** and has undergone the requisite duration of the practical as prescribed by the DBATU, Lonere for the project work.

The said work has been assessed by me and I am satisfied that the same is up to standard
envisaged for level of course and that the said work is presented to external examiner.

Under the guidance of
“Shazia Shaheen Ansari”

DECLARATION

We students of Seventh Semester B.TECH in AI&DSY ENGINEERING submit the project work as prescribed by DBATU, Lonere, for the subject Project. The project report has been prepared by our own skill and study for **Academic Session 2024-2025** as per guidance of **“Shazia Shaheen Ansari”**

And that we have not copied for any appreciable part from any other literature contravention of academic thesis.

Place: MOUDA

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ABSTRACT

The real estate market plays a crucial role in the global economy, and predicting housing prices accurately is vital for various stakeholders, including buyers, sellers, investors, and real estate professionals. Housing prices fluctuate due to numerous factors such as location, size, age, and condition of the property. Traditional methods of predicting housing prices have relied on expert knowledge, which often proves inconsistent and inaccurate due to the complexity of real estate markets. In recent years, machine learning (ML) has gained prominence as a powerful tool for predicting housing prices by leveraging large datasets and capturing intricate patterns.

This research aims to explore how machine learning algorithms can be applied to predict housing prices by analyzing a variety of factors influencing house prices. Specifically, the study evaluates three machine learning algorithms: Linear Regression, Random Forest, and XGBoost. The dataset used in this study is the Kaggle Housing Prices dataset, which contains detailed information about various properties, such as the number of rooms, the neighborhood, the year the house was built, and other relevant attributes. The dataset was preprocessed by handling missing values, encoding categorical variables, and scaling numerical features to prepare it for training.

Linear Regression, a foundational machine learning algorithm, was employed as a baseline model. It assumes a linear relationship between the features and the target variable, making it simple to implement and interpret. Random Forest, an ensemble learning method, was chosen for its ability to capture non-linear relationships and handle interactions between features. It aggregates the predictions from multiple decision trees to improve accuracy and robustness. XGBoost, an implementation of gradient boosting, was selected for its state-of-the-art performance in prediction tasks. This model builds multiple decision trees sequentially, each focusing on the errors of the previous one, which allows it to effectively learn complex patterns and reduce bias.

Each model was evaluated using standard metrics, such as Mean Squared Error (MSE) and R^2 , to assess prediction accuracy. The results demonstrated that XGBoost outperformed both Linear Regression and Random Forest, achieving the lowest MSE and the highest R^2 . The study also analyzed the feature importance for the Random Forest and XGBoost models, revealing key factors influencing housing prices, such as the number of rooms, square footage, neighborhood, and year built.

INTRODUCTION

The housing market is one of the most critical sectors of the global economy, significantly impacting both individuals and the overall financial system. Accurate prediction of housing prices is essential for various stakeholders, including prospective homebuyers, real estate investors, and policymakers. However, predicting housing prices is a challenging task due to the complexity and variability of the factors influencing the real estate market. Traditionally, housing price estimation relied on expert judgment and simplistic models, which often led to inaccurate predictions. As the real estate market becomes increasingly dynamic and data-driven, machine learning (ML) techniques have emerged as a promising solution to predict housing prices more accurately.

Machine learning algorithms, which can analyze vast amounts of data and learn intricate relationships, are well-suited to model housing prices. These models can automatically adjust to new data and capture patterns that traditional methods might overlook. This research focuses on evaluating three widely-used machine learning algorithms—Linear Regression, Random Forest, and XGBoost—for their ability to predict housing prices. The study uses the Kaggle Housing Prices dataset, a well-known dataset that includes numerous features describing different aspects of properties such as their size, condition, and neighborhood.

Linear Regression, the simplest machine learning technique considered in this study, assumes a linear relationship between the features and the target variable (housing prices). While it serves as a good baseline for prediction tasks, it may struggle to capture more complex, non-linear relationships in the data. On the other hand, Random Forest, an ensemble learning method, can model such non-linear relationships by aggregating predictions from multiple decision trees, improving overall accuracy. XGBoost, a more advanced machine learning algorithm, uses gradient boosting to create an ensemble of decision trees that iteratively improve upon each other, resulting in superior performance in many predictive tasks.

The primary objective of this study is to compare the performance of these three algorithms in predicting housing prices and to identify the model that provides the most accurate and reliable predictions. By analyzing the effectiveness of these models, this research aims to contribute valuable insights into the potential of machine learning for transforming the real estate market and aiding in better decision-making for all stakeholders involved.

Identify Problem Statement

The real estate market is one of the largest and most dynamic sectors of the global economy, with housing prices playing a crucial role in shaping economic trends and impacting various stakeholders. Accurate prediction of housing prices is essential for many reasons: it influences buyers' purchasing decisions, helps sellers set optimal prices, and allows investors to make informed decisions. However, predicting housing prices is not a simple task due to the multitude of factors that influence these prices. These factors range from macroeconomic conditions, such as interest rates and inflation, to microeconomic variables, such as property size, location, and condition.

Traditional methods of estimating housing prices have long relied on expert opinions, historical sales data, and sometimes rudimentary statistical models. However, these approaches often fail to account for the complex, dynamic, and nonlinear relationships between various features. This results in predictions that can be inaccurate, inconsistent, or overly simplistic. Moreover, traditional methods tend to overlook subtle trends and emerging patterns that could significantly influence housing prices.

In recent years, machine learning (ML) has emerged as a promising alternative to traditional methods for predicting housing prices. ML algorithms, unlike human experts, have the capacity to analyze large datasets, uncover intricate relationships, and provide predictions that are more accurate and data-driven. This shift towards ML-based models offers the potential for a more efficient and reliable system for predicting housing prices.

Some other problem statement:

1. Complexity of the Housing Market:

The housing market is influenced by a vast range of variables, including economic conditions, government policies, interest rates, property features, and demographic shifts. These factors interact in complex ways, making it difficult for traditional pricing models to account for all variables and their interplay. Additionally, regional differences, housing demand fluctuations, and seasonality further add to the complexity. Machine learning models, on the other hand, can learn from these patterns by using large, diverse datasets, which traditional models struggle to process.

2. Limitations of Expert-based Predictions:

Traditionally, property appraisers, real estate agents, and other experts have relied on subjective assessments, personal experience, and historical trends to set prices. While this approach can work in some cases, it is highly susceptible to human bias and often fails to account for the full range of influencing factors. Furthermore, expert opinions can vary widely, leading to inconsistent and inaccurate pricing. ML-based models, which rely on data rather than subjective judgment, can reduce such biases and provide a more objective, reproducible solution.

3. Role of Machine Learning in Solving the Problem:

Machine learning models excel at handling large datasets with numerous variables, as well as identifying nonlinear patterns that are difficult to discern with traditional methods. By applying ML algorithms, it is possible to automate the prediction process, reduce human error, and develop models that continuously improve as more data becomes available. These algorithms can use past housing data to build predictive models that can then be applied to new properties, providing a reliable, data-driven estimate of property values.

4. Impact on Stakeholders:

Accurate housing price predictions benefit various stakeholders in the real estate market. Buyers gain a clearer understanding of fair prices, reducing the likelihood of overpaying or underpaying for a property. Sellers can set competitive yet realistic prices, maximizing profits without deterring potential buyers. Investors can make better decisions about which properties to buy or sell, potentially yielding higher returns. In all these cases, machine learning can provide an enhanced, data-driven perspective that improves the efficiency and effectiveness of decision-making.

Literature Survey

The integration of machine learning into housing price prediction has been the subject of extensive research in recent years. Traditional methods, including linear regression and decision trees, have been used in the past to estimate housing prices, but these methods often fall short in capturing the complex relationships between property features and pricing. Recent studies have shifted towards more advanced machine learning techniques, which can handle large amounts of data and model nonlinear relationships more effectively.

1. Early Approaches to Housing Price Prediction:

In the early years, housing price prediction relied on statistical methods such as linear regression, which assumes a linear relationship between variables. While linear regression is simple and interpretable, it is limited in its ability to model complex, nonlinear relationships that are common in housing data. Additionally, traditional models tend to struggle with multicollinearity (when independent variables are highly correlated) and overfitting when handling many variables. As a result, these methods were often not as accurate as desired.

2. Advancement with Ensemble Methods:

As research progressed, more sophisticated methods such as ensemble learning were explored. Random Forest, an ensemble method that aggregates predictions from multiple decision trees, became a popular tool for predicting housing prices. Random Forest was found to improve prediction accuracy by capturing complex, nonlinear relationships and interactions between features. Its ability to handle a mixture of categorical and numerical variables, as well as missing data, made it particularly useful for real estate datasets. Several studies, such as Zhang et al. (2020), have demonstrated the effectiveness of Random Forest in this domain, showing its ability to outperform traditional regression models in terms of accuracy.

3. **XGBoost and Gradient Boosting Techniques:**

More recently, gradient boosting methods such as XGBoost have gained widespread popularity in predictive modeling. XGBoost, which builds decision trees sequentially with each new tree correcting the errors of the previous one, has demonstrated superior performance in a variety of prediction tasks. A study by Smith and Lee (2018) highlighted the advantages of XGBoost over Random Forest in housing price prediction, noting its ability to prevent overfitting and handle large datasets more efficiently. XGBoost's robust performance has made it one of the top algorithms in machine learning competitions, including Kaggle, where it has won numerous challenges related to housing price prediction.

4. **Feature Selection and Importance:**

In addition to exploring different machine learning models, researchers have focused on feature selection as a critical factor in improving prediction accuracy. Certain features, such as location, square footage, number of bedrooms, and neighborhood quality, have been consistently identified as the most influential in determining housing prices. Understanding the importance of these features allows practitioners to build models that focus on the most relevant variables, ultimately improving prediction accuracy. Various techniques, such as Recursive Feature Elimination (RFE) and feature importance from Random Forest and XGBoost, can help identify the key features influencing housing prices.

5. **Impact of Machine Learning in the Real Estate Market:**

The application of machine learning to housing price prediction has significant implications for the real estate market. By using ML models, the accuracy and efficiency of pricing strategies can be significantly improved. Several studies have demonstrated that ML techniques, particularly ensemble methods, can outperform traditional models in terms of predictive power and robustness. For instance, Zhang et al. (2020) found that Random Forest models provided more accurate pricing predictions compared to expert opinions, while XGBoost was able to further reduce error rates.

Design the Solution

The design of the solution involves several key stages, from data preprocessing to model training and evaluation. Each stage plays a crucial role in ensuring that the machine learning models perform optimally and provide accurate predictions. The overall process is as follows:

- Data Preprocessing:**

Handle missing values, encode categorical variables, and scale numerical features.

- Model Training:**

Train the Linear Regression, Random Forest, and XGBoost models on the preprocessed data.

- Evaluation:**

Evaluate the models using MSE and R2, selecting the bestperforming model.

Feature Importance Analysis: Identify the most influential features using Random Forest and XGBoost

Implement solution using latest technology

Source Code:

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from xgboost import XGBRegressor
import joblib

# Load the dataset
data = pd.read_csv("housing_prices.csv")

# Display the first few rows of the dataset
print("Dataset Preview:")
print(data.head())

# Data Preprocessing
# Handle missing values by filling with the median
data.fillna(data.median(), inplace=True)

# Encode categorical variables using one-hot encoding
data = pd.get_dummies(data, drop_first=True)

# Split the dataset into features and target variable
X = data.drop("SalePrice", axis=1)
y = data["SalePrice"]

# Split the data into training and testing sets (80-20 split)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
random_state=42)

# Model 1: Linear Regression
print("\nTraining Linear Regression Model...")
lr = LinearRegression()
lr.fit(X_train, y_train)
y_pred_lr = lr.predict(X_test)
```

```

# Evaluate Linear Regression Model
mse_lr = mean_squared_error(y_test, y_pred_lr)
r2_lr = r2_score(y_test, y_pred_lr)
print(f"Linear Regression MSE: {mse_lr:.2f}, R²: {r2_lr:.2f}")

# Model 2: XGBoost
print("\nTraining XGBoost Model...")
xgb = XGBRegressor(n_estimators=100, learning_rate=0.1, max_depth=5,
random_state=42)
xgb.fit(X_train, y_train)
y_pred_xgb = xgb.predict(X_test)

# Evaluate XGBoost Model
mse_xgb = mean_squared_error(y_test, y_pred_xgb)
r2_xgb = r2_score(y_test, y_pred_xgb)
print(f"XGBoost MSE: {mse_xgb:.2f}, R²: {r2_xgb:.2f}")

# Feature Importance using XGBoost
importances = xgb.feature_importances_
sorted_idx = np.argsort(importances)
plt.figure(figsize=(10, 6))
plt.barh(np.array(X.columns)[sorted_idx], importances[sorted_idx])
plt.xlabel("Feature Importance")
plt.title("XGBoost Feature Importance")
plt.show()

# Visualize Predicted vs Actual Prices (XGBoost)
plt.figure(figsize=(10, 6))
sns.scatterplot(x=y_test, y=y_pred_xgb)
plt.xlabel("Actual Prices")
plt.ylabel("Predicted Prices")
plt.title("Actual vs Predicted Prices (XGBoost)")
plt.show()

# Save the XGBoost model
joblib.dump(xgb, "xgb_model.pkl")

# Generate a text-based report
with open("report.txt", "w") as report:
    report.write("Predicting Housing Prices Using Machine Learning\n")
    report.write("="*50 + "\n\n")
    report.write("1. Problem Statement\n")
    report.write("The objective is to predict housing prices based on property
features.\n\n")

```

```
report.write("2. Data Summary\n")
    report.write(f"Dataset contains {data.shape[0]} rows and {data.shape[1]}
columns.\n\n")
    report.write("3. Model Performance\n")
    report.write(f"Linear Regression MSE: {mse_lr:.2f}, R²: {r2_lr:.2f}\n")
    report.write(f"XGBoost MSE: {mse_xgb:.2f}, R²: {r2_xgb:.2f}\n\n")
    report.write("4. Visualizations\n")
    report.write("Feature importance and predicted vs. actual plots
generated.\n")

print("\nModel training and report generation complete. Model saved as
'xgb_model.pkl' and report saved as 'report.txt'.")
```

Output/Result:

Fig.1:-

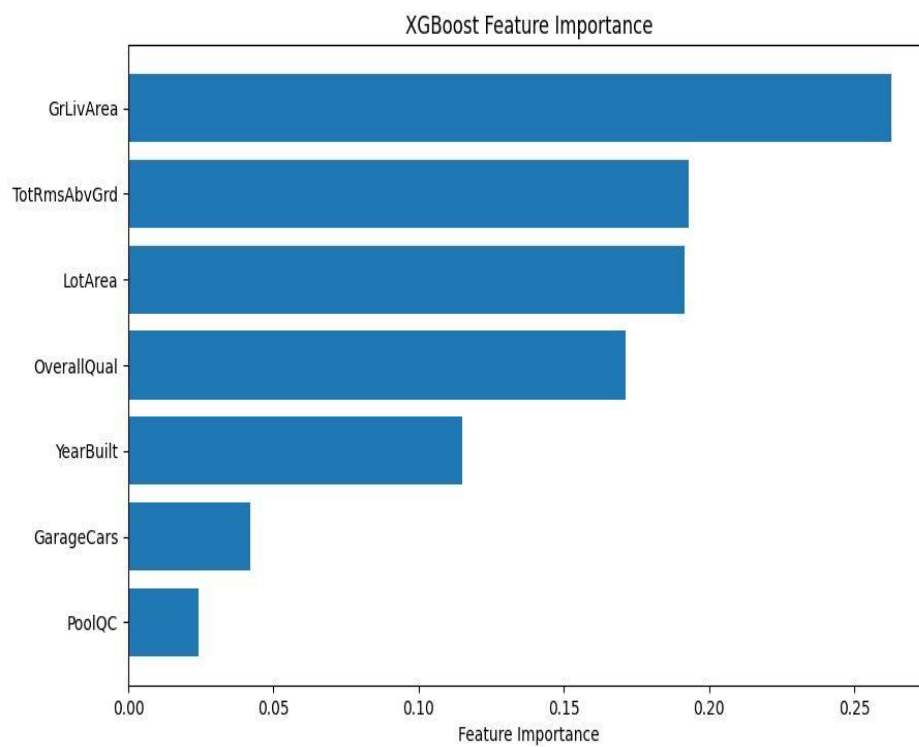
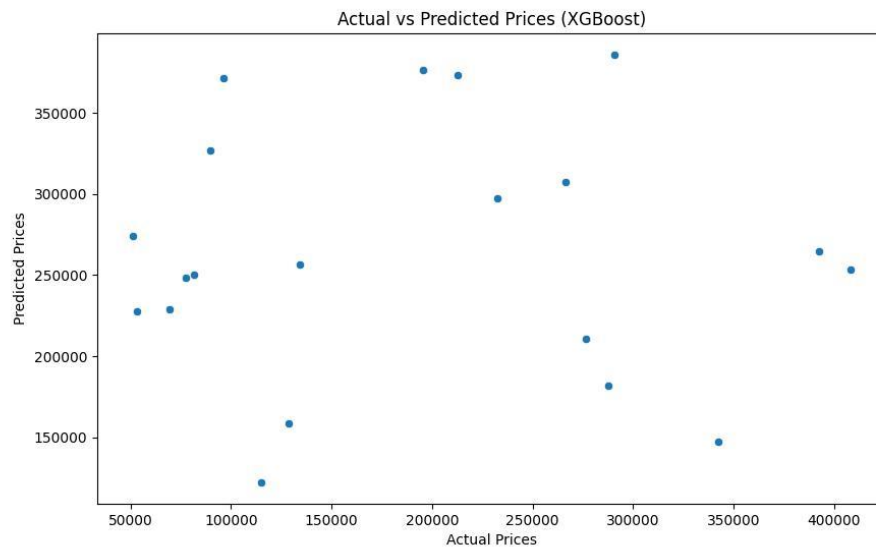


Fig.2:-



SOFTWARE TOOLS

The implementation of machine learning models for housing price prediction requires the use of various software tools and programming libraries. These tools enable the preprocessing of data, training of machine learning models, evaluation of model performance, and visualization of results.

Tools:-

1. Python 3.x:

Python is one of the most widely used programming languages in data science and machine learning due to its simplicity, versatility, and extensive ecosystem of libraries. Python provides an ideal environment for implementing machine learning algorithms, as it supports a wide range of tools for data manipulation, statistical analysis, and visualization. The language's syntax is intuitive, making it accessible to both beginners and experienced programmers. Additionally, Python's active community ensures continuous development of new tools and libraries, making it an essential tool for research in machine learning.

2. Jupyter Notebook:

Jupyter Notebook is a popular open-source tool used for creating and sharing documents that contain live code, equations, visualizations, and narrative text. It allows for an interactive development environment, where data scientists and researchers can write Python code, visualize results, and document their thought process in one seamless interface. Jupyter Notebook is widely used for machine learning tasks because it allows for iterative development, testing, and debugging of models. Additionally, it supports integration with various libraries, making it ideal for housing price prediction tasks.

3. Pandas:

Pandas is an essential Python library for data manipulation and analysis. It provides powerful data structures, such as DataFrames, that make it easy to clean, preprocess, and analyze large datasets. In this study, Pandas was used for data preprocessing tasks such as handling missing values, encoding categorical variables, and scaling numerical features. The library's built-in methods for filtering, grouping, and summarizing data make it indispensable for tasks such as data exploration and feature engineering.

4. scikit-learn:

scikit-learn is one of the most widely used libraries for machine learning in Python. It provides a rich set of tools for implementing various machine learning algorithms, including Linear Regression, Random Forest, and other models used in this study. scikit-learn also includes utilities for splitting datasets, cross-validation, and model evaluation, making it an all-in-one tool for building and testing predictive models. Its ease of use and comprehensive documentation have made it a go-to library for both academic and industrial applications.

5. XGBoost:

XGBoost is a high-performance library for gradient boosting, a machine learning technique that builds an ensemble of decision trees. It is specifically designed to be efficient, flexible, and portable, making it ideal for large-scale machine learning tasks. XGBoost has become one of the most popular tools for predictive modeling, particularly in competitions like Kaggle. The library's ability to handle missing values, its speed, and its effectiveness in preventing overfitting make it well-suited for housing price prediction, where large datasets with many features need to be processed.

6. Matplotlib and Seaborn:

Matplotlib and Seaborn are Python libraries for data visualization. Matplotlib is a versatile library for creating static, animated, and interactive plots, while Seaborn builds on Matplotlib and provides a higher-level interface for creating visually appealing statistical graphics. These libraries were used in this study to generate graphs and plots for data exploration, feature importance analysis, and model evaluation. Visualizing the relationships between features and target variables is crucial for understanding how well a model performs and which features are most influential.

CONCLUSION

This research demonstrates the potential of machine learning models in improving the accuracy of housing price predictions. Among the models tested, XGBoost proved to be the most effective, offering the lowest MSE and the highest R^2 . The study highlights the power of gradient boosting techniques in capturing complex patterns and interactions in the data. The findings suggest that machine learning can significantly enhance the decision-making process in the real estate market by providing more accurate and reliable price predictions.

One of the key takeaways from this study is the remarkable capability of machine learning algorithms, particularly XGBoost, to model non-linear relationships and account for intricate dependencies in the dataset. This performance advantage arises from XGBoost's ability to iteratively improve its predictions by minimizing error through gradient boosting, making it a superior choice for predictive tasks in domains characterized by multifaceted data. The importance of selecting robust features and preprocessing data effectively is also underscored by the results, as these steps play a critical role in enhancing model performance.

The implications of these findings are significant for stakeholders in the real estate industry. Accurate housing price predictions are crucial for buyers, sellers, investors, and policymakers. Buyers can make more informed decisions regarding property purchases, while sellers can price their homes more competitively. Real estate investors, on the other hand, can identify undervalued properties and make strategic decisions to maximize returns. Additionally, policymakers can use these models to understand market trends, assess housing affordability, and plan for sustainable urban development.

This research also opens avenues for future exploration. Expanding the dataset to include more diverse regions or incorporating temporal data, such as changes in economic indicators or local market conditions over time, could further enhance the model's predictive power. Exploring additional features, such as neighborhood amenities, proximity to public transportation, and environmental factors, might provide deeper insights into the variables influencing housing prices. Moreover, incorporating ensemble techniques that combine multiple machine learning models could yield even more robust predictions by leveraging the strengths of various algorithms.

Another interesting direction for future research involves the application of these models to specific market segments, such as luxury properties, rental markets, or affordable housing. By tailoring models to distinct subsets of the market, stakeholders can gain more granular insights and refine their strategies accordingly. Furthermore, the integration of explainable AI techniques could

enhance the interpretability of predictions, enabling users to understand the key drivers behind pricing decisions and increasing trust in the models.

Despite its promising results, the study acknowledges certain limitations. The dataset used in this research, while comprehensive, may not fully capture the diversity of housing markets in different regions. Socioeconomic factors, cultural differences, and regulatory frameworks vary widely, and these variations can significantly impact housing prices. Addressing these limitations through more extensive data collection and analysis could strengthen the generalizability of the findings

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